CLA 2 QB with Answers

Chemistry (SRM Institute of Science and Technology)

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Department of Chemistry

Even semester 2018-19

Question Bank [4/6/8/10/12 marks] CLA 2 Portions only

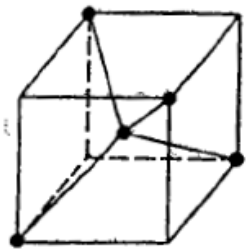
MODULE II

1. Discuss the crystal field splitting in tetrahedral complexes.

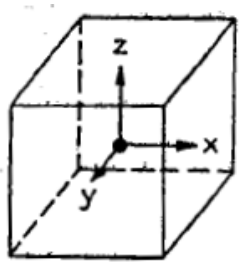
(OR)

Draw and discuss the energy level diagram for splitting of d orbitals in a Td ligand field taking an example.

A regular tetrahedron is related to a cube. One atom is at the centre of the cube, and four of the eight corners of the cube are occupied by ligands



The directions x, y and z point to the centres of the faces of the cube.



The e_g orbitals point along x, y and z (that is to the centres of the faces). The t_{2g} orbitals point between x, y and z (that is towards the centres of the edges of the cube) (Figure 7.21).

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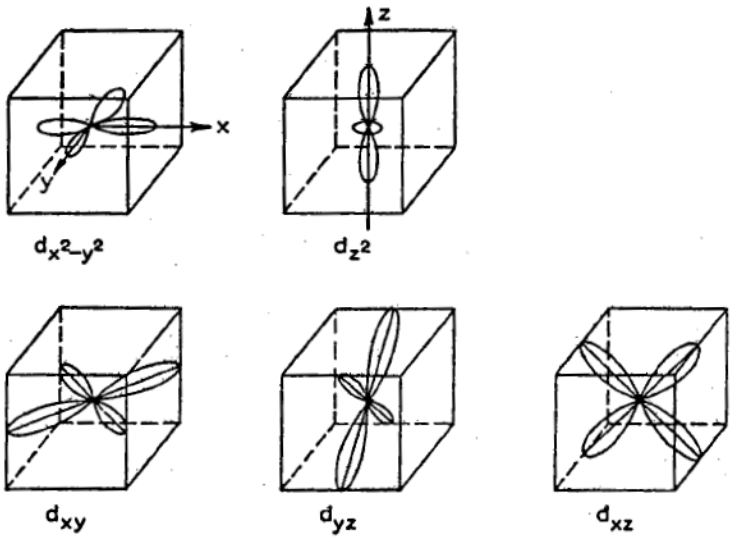
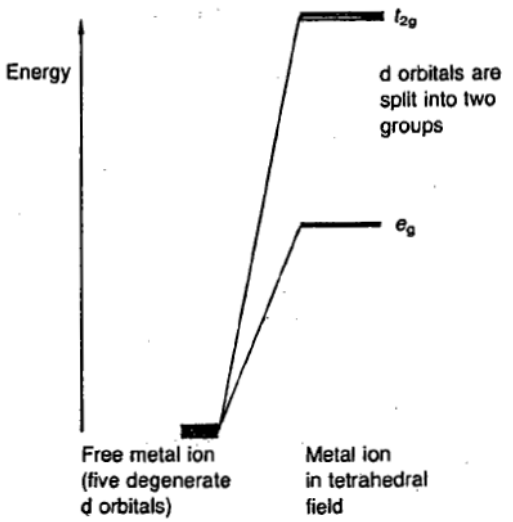


Figure 7.21 Orientation of *d* orbitals relative to a cube.

The direction of approach of the ligands does not coincide exactly with either the e_g or the t_{2g} orbitals. The angle between an e_g orbital, the central metal and the ligand is half the tetrahedral angle = 109°28′/2 = 54°44′. The angle between a t_{2g} orbital, the central metal and the ligand is 35°16′. Thus the t_{2g} orbitals are nearer to the direction of the ligands than the e_g orbitals. (Alternatively the t_{2g} orbitals are half the side of the cube away from the approach of the ligands, whilst the e_g orbitals are half the diagonal of the cube away.)

The approach of the ligands raises the energy of both sets of orbitals. The energy of the t_{2g} orbitals is raised most because they are closest to the ligands.

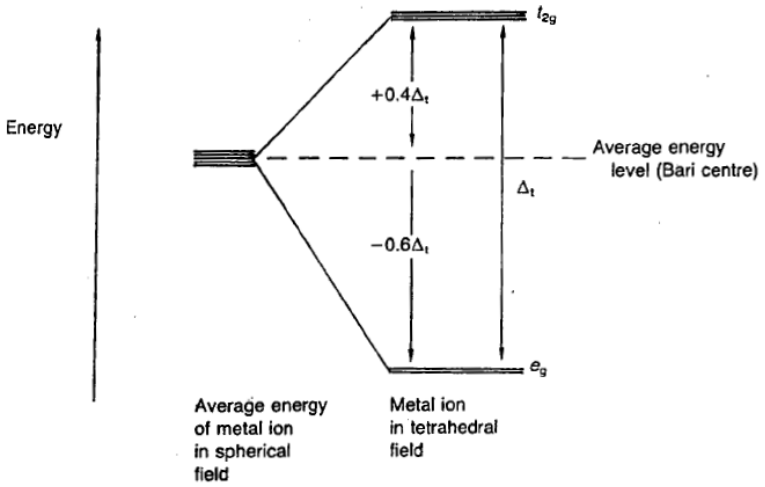


Crystal field splitting of energy levels in a tetrahedral field.

This crystal field splitting is the opposite way round to that in octahedral complexes

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The t_{2g} orbitals are $0.4\Delta_t$ above the weighted average energy of the two groups (the Bari centre) and the e_g orbitals are $0.6\Delta_t$ below the average



The magnitude of the crystal field splitting Δ_t in tetrahedral complexes is considerably less than in octahedral fields. There are two reasons for this:

1. There are only four ligands instead of six, so the ligand field is only two thirds the size: hence the ligand field splitting is also two thirds the size.
2. The direction of the orbitals does not coincide with the direction of the ligands. This reduces the crystal field splitting by roughly a further two thirds.

Thus the tetrahedral crystal field splitting Δ_t is roughly $2/3 \times 2/3 = 4/9$ of the octahedral crystal field splitting Δ_o. Strong field ligands cause a bigger energy difference between t_{2g} and e_g than weak field ligands. However, the tetrahedral splitting Δ_t is always much smaller than the octahedral splitting Δ_o. Thus it is never energetically favourable to pair electrons, and all tetrahedral complexes are high-spin.

2. What is a spectrochemical series? Mention its importance.

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Examination of the spectra of a series of complexes of the same metal with different ligands shows that the position of the absorption band (and hence the value of Δ_o) varies depending on the ligands which are attached (Table 7.8).

Ligands which cause only a small degree of crystal field splitting are termed weak field ligands. Ligands which cause a large splitting are called strong field ligands. Most Δ values are in the range $7000\,cm^{-1}$ to $30000\,cm^{-1}$. The common ligands can be arranged in ascending order of crystal field splitting Δ. The order remains practically constant for different metals, and this series is called the spectrochemical series (see Further Reading Tsuchida, 1938; Jørgensen, 1962).

Spectrochemical series

weak field ligands

$I^- < Br^- < S^{2-} < Cl^- < NO_3^- < F^- < OH^- <$ EtOH < oxalate < H_2O < EDTA < (NH_3 and pyridine) < ethylenediamine < dipyridyl < o-phenanthroline < NO_2^- < CN^- < CO

strong field ligands

3. Why Td complexes are mostly high spin complexes rather low spin?

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4. Calculate CFSE for d^3, d^5, d^6, $d^7$and d^8 configurations for tetrahedral complexes.

$CFSE_{(Tetrahedral)} = -0.6xn_{eg} + 0.4xn_{t2g}$

n_{eg} is number of electrons in e_g orbital and n_{t2g} is number of electrons in t_{2g} orbital.

Number of *d* electrons	Arrangement of electrons	Spin only magnetic moment	Tetrahedral CFSE

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	e_g	t_{2g}	μ(D)	Δ_t
d^3	↑ ↑	↑	3.87	−1.2 + 0.4 = −0.8
d_5	↑ ↑	↑ ↑ ↑	5.92	−1.2 + 1.2 = 0.0
d^6	↑↓ ↑	↑ ↑ ↑	4.90	−1.8 + 1.2 = −0.6
d_7	↑↓ ↑↓	↑ ↑ ↑	3.87	−2.4 + 1.2 = −1.2
d^8	↑↓ ↑↓	↑↓ ↑ ↑	2.83	−2.4 + 1.6 = −0.8

5. Give the differences between diamagnetic and paramagnetic complexes.

- Paramagnetic materials are attracted by external magnetic fields whereas diamagnetic materials are repelled.
- Paramagnetic materials have at least one unpaired electron in the system, but diamagnetic materials have all their electrons paired.
- The magnetic field created by paramagnetic materials are in the direction of the external magnetic field whereas the magnetic field created by diamagnetic materials are opposing in direction to the external magnetic field.
- Paramagnetism is a stronger magnetic behaviour exhibited only by selective materials, whereas diamagnetism is a weak magnetic behaviour generally shown by all materials and easily suppressed in the presence of stronger magnetic properties.

6. Explain the terms a. Ferromagnetism b. Anti-ferromagnetism and c. Ferrimagnetism

Ferromagnetic ↑ ↑ ↑ ↑ ↑ ↑ ↑	Below T_C, spins are aligned parallel in magnetic domains
Antiferromagnetic ↑ ↓ ↑ ↓ ↑ ↓ ↑	Below T_N, spins are aligned antiparallel in magnetic domains
Ferrimagnetic ↑ ↓ ↑ ↓ ↑ ↓ ↑	Below T_C, spins are aligned antiparallel but do not cancel
Paramagnetic ↑ ↓ ↑ ↑ ↓ ↑ ↓	Spins are randomly oriented (any of the others above T_C or T_N)

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ferromagnetic ordering, in which all the spins within a magnetic domain have the same orientation, as shown in the figure. Conversely, an antiparallel alignment of spins between neighboring atoms results in **antiferromagnetic** ordering. A third arrangement, **ferrimagnetic** ordering, results from an antiparallel alignment of spins on neighboring atoms when the magnetic moments of the neighbors are unequal. In this case, the spin moments do not cancel and there is a net magnetization. The ordering mechanism is like that of an antiferromagnetic solid, but the magnetic properties resemble those of a ferromagnet. Ferrimagnetic ordering is most common in metal oxides.

7. Explain the magnetic properties of metal complexes with suitable examples.

Magnetic properties arise from the spin and orbital angular momentum of the electrons contained in a compound. Compounds are diamagnetic when they contain no unpaired electrons. Molecular compounds that contain one or more unpaired electrons are paramagnetic. The magnitude of the paramagnetism is expressed as an effective magnetic moment, μ_{eff}.

For first-row transition metals the magnitude of μ_{eff} is, to a first approximation, a simple function of the number of unpaired electrons, the spin-only formula. In general, spin-orbit coupling causes μ_{eff} to deviate from the spin-only formula. For the heavier transition metals, lanthanides and actinides, spin-orbit coupling cannot be ignored.

relative orientations of the spins of nearby domains resulting in ferromagnetism, antiferromagnetism or ferrimagnetism (**Explanation from earlier question**).

When orbital angular momentum is completely quenched, $\vec{L} = 0$ and the paramagnetism can be attributed to electron spin alone. The total spin angular momentum is simply half the number of unpaired electrons and the spin-only formula results.

$$\mu_{\text{eff}} = \sqrt{n(n+2)}\mu_B$$

where n is the number of unpaired electrons.

μ_B is Bohr magneton, the unit for magnetic moment.

Ion	Number of unpaired electrons	Spin-only moment /μ_B	observed moment /μ_B
Ti^{3+}	1	1.73	1.73
V^{4+}	1	1.73	1.68–1.78
Cu^{2+}	1	1.73	1.70–2.20
V^{3+}	2	2.83	2.75–2.85
Ni^{2+}	2	2.83	2.8–3.5
V^{2+}	3	3.87	3.80–3.90
Cr^{3+}	3	3.87	3.70–3.90
Co^{2+}	3	3.87	4.3–5.0
Mn^{4+}	3	3.87	3.80–4.0
Cr^{2+}	4	4.90	4.75–4.90
Fe^{2+}	4	4.90	5.1–5.7
Mn^{2+}	5	5.92	5.65–6.10
Fe^{3+}	5	5.92	5.7–6.0

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8. For each of the following coordination complexes, identify whether it is paramagnetic or diamagnetic based on magnetic moment values?

   a. Octahedral-low spin-d^4 b. **octahedral- high spin-d^6** c. **tetrahedral- high spin-d^7**.

Calculate the magnetic moment value according to the following equation and according to the number of unpaired electron in each condition.

$$\mu_{\text{eff}} = \sqrt{n(n+2)}\mu_B$$

where *n* is the number of unpaired electrons.

S.No.	Configuration	Number of unpaired electrons	μ_{eff}	Paramagnetic or diamagnetic
a	low spin-d^4 (t_{2g}^4)	2	2.83	Paramagnetic
b	high spin-d^6 ($t_{2g}^4 e_g^2$)	4	4.90	Paramagnetic
c	high spin-d^7 ($t_{2g}^5 e_g^2$)	3	3.87	Paramagnetic

9. Identify from the following as high spin or low spin complexes and calculate the magnetic moment of the complexes.

i. $[CoF_6]^{3-}$ ii. $[NiCl_4]^{2-}$ iii. $[Fe(CN)_6]^{3-}$ iv. $[Co(H_2O)_6]^{3+}$ v. $[FeCl_4]^-$

$[CoF_6]^{3-}$: high spin as weak ligand, d^6, t_{2g}^6, 4 unpaired electrons, μ_{eff} is 4.90.

$[NiCl_4]^{2-}$: tetrahedral, high spin as weak ligand, d^8 ($e^4 t_2^4$), 2 unpaired electrons, μ_{eff} is 2.83.

$[Fe(CN)_6]^{3-}$: low spin as strong ligand, d^5, t_{2g}^5, 1 unpaired electron, μ_{eff} is 1.73.

$[Co(H_2O)_6]^{3+}$: low spin as strong ligand, d^6, t_{2g}^6, 0 unpaired electron, μ_{eff} is 0.

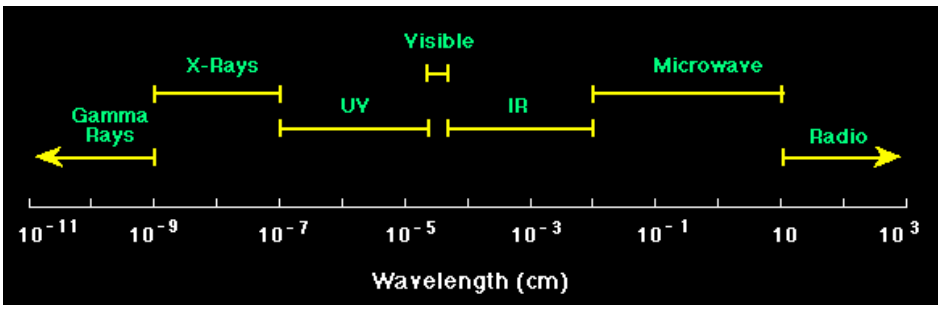
$[FeCl_4]^-$: high spin as weak ligand, d^6, e^3 $t2^3$, 4 unpaired electrons, μ_{eff} is 4.90.

10. What are the different regions of electromagnetic radiations?

The **electromagnetic spectrum** is generally divided into seven **regions**, in order of decreasing wavelength and increasing **energy** and frequency: radio **waves**, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays.

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11. Explain the characteristics of Electromagnetic radiation?

Electromagnetic radiation is a form of energy that is produced by oscillating electric and magnetic disturbance, or by the movement of electrically charged particles traveling through a vacuum or matter. The electric and magnetic fields come at right angles to each other and combined wave moves perpendicular to both magnetic and electric oscillating fields thus the disturbance. Electron radiation is released as photons, which are bundles of light energy that travel at the speed of light as quantized harmonic waves. This energy is then grouped into categories based on its wavelength into the electromagnetic spectrum. These electric and magnetic waves travel perpendicular to each other and have certain characteristics, including amplitude, wavelength, and frequency.

General Properties of all electromagnetic radiation:

1. Electromagnetic radiation can travel through empty space. Most other types of waves must travel through some sort of substance. For example, sound waves need either a gas, solid, or liquid to pass through in order to be heard.
2. The speed of light is always a constant. (Speed of light : 2.99792458×10^{8} m s^{-1})
3. Wavelengths are measured between the distances of either crests or troughs. It is usually characterized by the Greek symbol λ

Characteristics of radiation are,

Amplitude

Amplitude is the distance from the maximum vertical displacement of the wave to the middle of the wave. This measures the magnitude of oscillation of a particular wave. In short, the amplitude is basically the height of the wave. Larger amplitude means higher energy and lower amplitude means lower energy. Amplitude is important because it tells you the intensity or brightness of a wave in comparison with other waves.

Wavelength

Wavelength (λ) is the distance of one full cycle of the oscillation. Longer wavelength waves such as radio waves carry low energy; Shorter wavelength waves such as x-rays carry higher. This wavelength frequently relationship is characterized by:

$$c = \lambda v$$

where

- c is the speed of light,
- λλ is wavelength, and
- vv is frequency.

Frequency

Frequency is defined as the number of cycles per second, and is expressed as sec^{-1} or Hertz (Hz). Frequency is directly proportional to energy and can be express as:

$$E = hv$$

where

- E is energy,
- h is Planck's constant, ($h= 6.62607 \times 10^{-34}$ J), and

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- v is frequency.

12. What is the significance of selection rule in spectroscopy?

Selection rules specify the possible transitions among quantum levels due to absorption or emission of electromagnetic radiation

In spectroscopy, a **forbidden mechanism** (**forbidden transition** or **forbidden line**) is a spectral line associated with absorption or emission of light by atomic nuclei, atoms, or molecules which undergo a transition that is not allowed by a particular selection rule but is allowed if the approximation associated with that rule is not made.

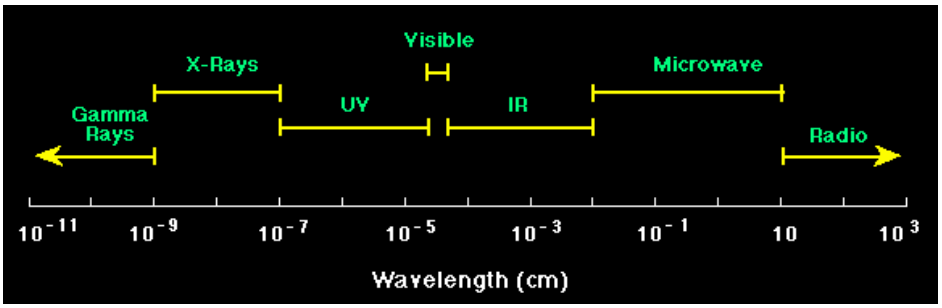
For example selection rule for rotational spectroscopy is $\Delta J = \pm 1$

Selection rule for vibrational spectroscopy is $\Delta v = \pm 1$

13. What is electromagnetic spectrum? Explain the different regions in it.

The **electromagnetic spectrum** is the range of frequencies of **electromagnetic** radiation and their respective wavelengths and energies.

The **electromagnetic spectrum** is generally divided into seven **regions**, in order of decreasing wavelength and increasing **energy** and frequency: radio **waves**, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays.



14. What are the criteria for a molecule to absorb in Microwave region? Give examples for microwave active and inactive molecules.

To be active in microwave spectroscopy, the molecule must possess a permanent dipole moment value.

Eg. For Microwave active molecule: heteronuclear diatomics - HCl, CO, NO, etc

Homonuclear diatomics are **microwave inactive** (e.g. O_2, N_2, etc.). For **CO_2** molecule, even though C-O bond is partially polar, net dipole moment = 0 as the molecule is linear; hence, it is **microwave** inactive.

15. Write notes on the selection rule for Rotational spectroscopy?

Selection rule for rotational spectroscopy is $\Delta J = \pm 1$, where J is rotational quantum number.

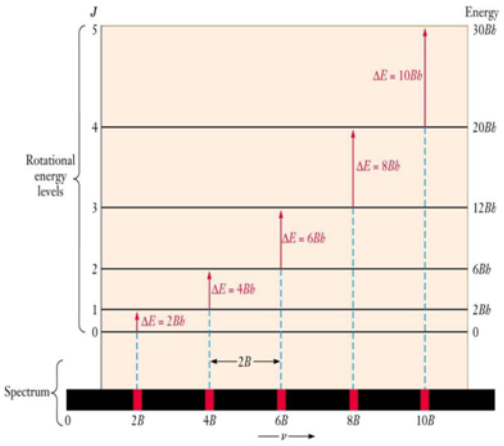
To raise the molecule from the state J to J+1, the wave number of radiation required is

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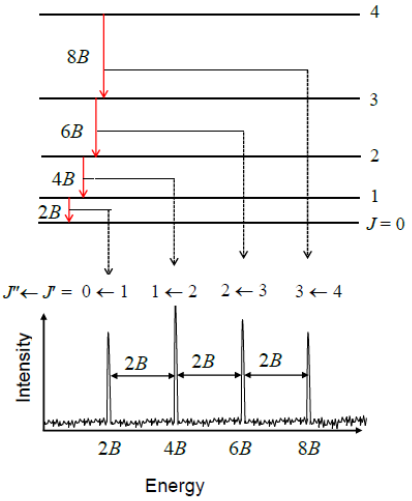
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$$\tilde{\nu}_{J \to J+1} = 2B(J+1) \quad \text{cm}^{-1}$$

Spectrum for ΔJ = +1



Spectrum for ΔJ = -1



16. What are allowed and forbidden transitions in spectroscopy? Explain with an example.

In spectroscopy, a **forbidden mechanism** (**forbidden transition** or **forbidden line**) is a spectral line associated with absorption or emission of light by atomic nuclei, atoms, or molecules which undergo a transition that is not allowed by a particular selection rule but is allowed if the approximation associated with that rule is not made.

For example selection rule for rotational spectroscopy is ΔJ = ±1, where J is rotational quantum number

If the transition occurs from J = 1 to J = 2, or J = 2 to J = 1, ie, $\Delta J = \pm 1$, the spectrum will contain lines for these transitions. But for J = 1 to J = 3 cannot be observed in the rotational spectra of any molecule. Therefore such transitions are called forbidden transitions.

17. Discuss in detail about the selection rule for rotational spectrum of a rigid diatomic molecule.

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Pure rotational spectra are observed in far infrared region. This spectra would be observed only with those molecules which are associated with a permanent electric dipole. A rotating diatomic molecule may be treated to a good approximation as a rigid rotator with an internuclear distance r.

Consider a diatomic molecule AB consisting of two atoms A and B of masses m_1 and m_2 respectively. Let the two atoms be joined by a rigid bond of length r. The two atoms A and B are at a distance of r_1 and r_2 from the centre of mass G of the molecule about which the molecule rotates end-over as shown in Fig. 25.2, so that

$$m_1 r_1 = m_2 r_2 \qquad ...(25.1)$$

and

$$r_1 + r_2 = r \qquad ...(25.2)$$

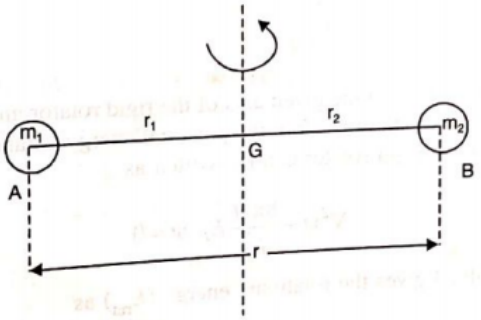


Fig. 25.2 Rigid rotator model for a diatomic molecule

The moment of inertia I with respect to a line (dotted as shown in Fig. 25.2) that passes through the centre of mass is

$$I = m_1 r_1^2 + m_2 r_2^2 \qquad ...(25.3)$$

Eliminating r_1 and r_2 between equation (25.1), (25.2) and (25.3) yields

$$I = m_1 \left(\frac{m_2}{m_1 + m_2} \right)^2 r^2 + m_2 \left(\frac{m_1}{m_1 + m_2} \right)^2 r^2$$

rearranging, we get

$$I = \frac{m_1 m_2}{(m_1 + m_2)^2} r^2 (m_1 + m_2)$$

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$$= \frac{m_1 m_2}{m_1 + m_2} r^2$$

$$= \mu r^2$$

where μ is the reduced mass of the diatomic molecule and is equal to $\frac{m_1 m_2}{m_1 + m_2}\left(\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}\right)$.

The rigid rotator has only kinetic energy and is given as

$$K.E. = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

where v_1 and v_2 are the linear velocities of mass m_1 and m_2 respectively. Since r_1 and r_2 are supposed to be constant (unchanged) during the rotation, therefore

$$K.E. = \frac{1}{2} m_1 (r_1 \omega)^2 + \frac{1}{2} m_2 (r_2 \omega)^2$$

$$= \frac{1}{2} I \omega^2$$

where I is the moment of inertia about the given axis of the rigid rotator and ω is the angular velocity.

Since no force acts on the rotator, therefore the potential energy is taken equal to zero, *i.e.*, $V = 0$. The Schrödinger equation for a rigid rotator can be written as

$$\nabla^2 \psi + \frac{8\pi^2 I}{h^2} E_{rot} \psi = 0$$

This equation when solved gives the rotational energy (E_{rot}) as

$$E_{rot} = \frac{h^2}{8\pi^2 I} J(J+1) \qquad ...(25.4)$$

where J is the rotational quantum number having integral values 0, 1, 2, ... and I is the moment of inertia of the molecule.

Since $E = hc\bar{\nu}$, therefore equation (25.4) can be written as

$$\bar{\nu} = \frac{h^2}{8\pi^2 Ihc} J(J+1)$$

$$= \frac{h}{8\pi^2 Ic} J(J+1) \qquad ...(25.5)$$

$$= BJ(J+1) \qquad ...(25.6)$$

where $B = \frac{h}{8\pi^2 Ic}$ is a constant and is generally referred to as the rotational constant and is expressed in cm^{-1} (or m^{-1}).

In order to predict the rotational spectrum, one needs to consider the difference between the rotational energy levels. The energy difference between two levels is given by

$$\Delta E_{rot} = E_{J'} - E_J$$

$$= \frac{h^2}{8\pi^2 I} J'(J'+1) - \frac{h^2}{8\pi^2 I} J(J+1) \qquad ...(25.7)$$

where J' is the higher quantum number and J is the lower quantum number ($J' > J$).

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$$\bar{\nu}_{J\to J'} = \frac{\Delta E_{rot}}{hc} = \frac{h^2}{8\pi^2 Ihc}\left[J'(J'+1) - J(J+1)\right]$$

$$= \frac{h}{8\pi^2 Ic}\left[J'(J'+1) - J(J+1)\right]$$

$$= B\left[J'(J'+1) - J(J+1)\right] \qquad ...(25.8)$$

When $J = 0$ and $J' = 1$, equation (25.8) becomes

$$\bar{\nu}_{0\to 1} = B\left[1(1+1) - 0(0+1)\right] = 2B$$

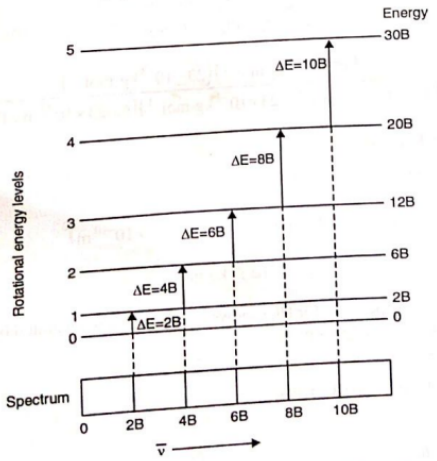


Fig. 25.3 Transitions between the various rotational levels and spectral lines arising from these transitions

Thus, for $J = 0$ and $J' = 1$ transition, an absorption line will appear at $2B$. Again, if the molecule is raised from $J = 1$ to $J' = 2$ level by the absorption of more energy in microwave region, then

$$\bar{\nu}_{1\to 2} = B\left[2(2+1) - 1(1+1)\right]$$

$$= 4B$$

It means that corresponding to this transition ($J = 1$ and $J' = 2$), an absorption line will appear at 4B. In general, when the molecule is raised from the level J to $J + 1$, the wave number for the transition is

$$\bar{\nu}_{J\to J+1} = B\left[(J+1)(J+2) - J(J+1)\right]$$

$$= B(J+1)[J+2-J]$$

$$= 2B(J+1) \qquad ...(25.9)$$

From equation (25.9) it is clear that stepwise, raising of rotational energy results in an absorption spectrum consisting of line at 2B, 4B, 6B, etc., with a constant separation of 2B. (Fig. 25.3).

In deriving this pattern we have made the assumption that a transition can occur from a particular level only to its immediate neighbour, either above or below: we have not, for instance, considered the sequence of transitions $J = 0 \to J = 2 \to J = 4 \ldots$ In fact, a rather sophisticated application of the Schrödinger wave equation shows that, for this molecule, we need only consider transitions in which J changes by one unit—all other transitions being spectroscopically *forbidden*. Such a result is called a *selection rule*, and we may formulate it for the rigid diatomic rotator as:

Selection rule: $\Delta J = \pm 1$

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18. Explain how Hooke's law is useful in comparing the vibrating bond to the physical model of a vibrating spring system.

(OR)

Explain the significance of Hooke's law in IR spectroscopy.

When two atoms combine to form a stable covalent molecule, for example HCl, they may be said to do so because of some internal electronic rearrangement. We shall not here discuss the detailed mechanisms of chemical bond formation, but we can simply look on the phenomenon as a balancing of forces. On the one hand there is a repulsion between the positively charged nuclei of both atoms, and between their negative electron 'clouds'; on the other there is an attraction between the nucleus of one atom and the electrons of the other, and vice versa. The two atoms settle at a mean internuclear distance such that these forces are just balanced and the total energy of the whole system is a minimum. Squeezing the atoms more closely together will cause the repulsive force to rise rapidly, while pulling them apart is resisted by the attractive force. Any attempt to distort the bond length requires an input of energy and we may plot energy against internuclear distance as in Fig. 3.1, where we have 'anchored' the chlorine atom on one axis and imagine pushing and pulling the hydrogen atom closer to or further from the chlorine—a bigger push or pull results in raising the energy more. At the energy minimum the internuclear distance is referred to as the equilibrium distance $r_{eq.}$, or more simply as the bond length.

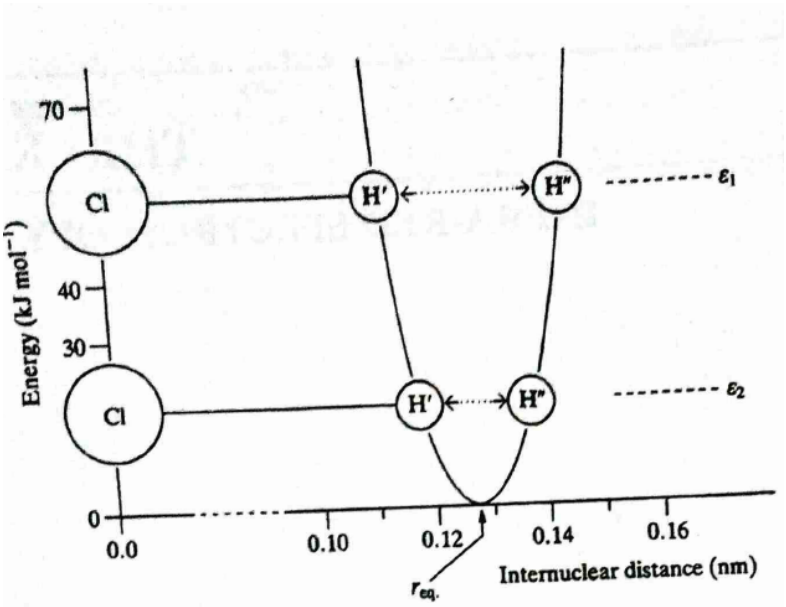


Figure 3.1 The energy of HCl as the bond is compressed or extended.

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The compression and extension of a bond may be likened to the behaviour of a spring, and we may extend the analogy by assuming that the bond, like a spring, obeys Hooke's law. We may write:

$$f = -k(r - r_{eq.}) \quad (3.1)$$

where f is the restoring force, k the force constant, and r the internuclear distance. In this case the energy curve is parabolic and has the form

$$E = \tfrac{1}{2}\, k(r - r_{eq.})^2 \quad (3.2)$$

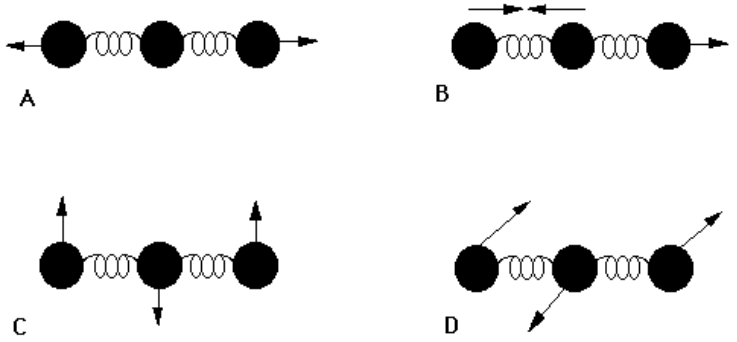
This model of a vibrating diatomic molecule—the so-called simple harmonic oscillator model—while only an approximation, forms an excellent starting point for the discussion of vibrational spectra.

19. What is the major requirement for infra-red absorption? [or] Explain the criteria for a molecule to absorb in IR region.

For a **molecule to absorb IR**, the vibrations within a **molecule** must cause a net change in the dipole moment of the **molecule**.

Carbon dioxide, a linear molecule, has 3 x 3 - 5 = 4 vibrations. These vibrational modes shown in Figure, are responsible for the "greenhouse" effect in which heat radiated from the earth is absorbed (trapped) by CO2 molecules in the atmosphere.

The arrows indicate the directions of motion. Vibrations labeled A and B represent the stretching of the chemical bonds, one in a symmetric (A) fashion, in which both C=O bonds lengthen and contract together (in-phase), and the other in an asymmetric (B) fashion, in which one bond shortens while the other lengthens. The asymmetric stretch (B) is infrared *active* because there is a change in the molecular dipole moment during this vibration. Infrared radiation at 2349 cm^{-1} (4.26 um) excites this particular vibration. The symmetric stretch is not infrared active, and so this vibration is not observed in the infrared spectrum of CO2. The two equal-energy bending vibrations in CO2 (C and D) are identical except that one bending mode is in the plane of the paper, and one is out of the plane. Infrared radiation at 667 cm^{-1} (15.00 um) excites these vibrations.



Vibrations of CO_2

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20. Write notes on the vibrational spectroscopy for a diatomic molecule behaving like simple harmonic oscillator.

(OR)

Explain the theory behind vibrational spectroscopy for a diatomic molecule.

When two atoms combine to form a stable covalent molecule, for example HCl, they may be said to do so because of some internal electronic rearrangement. We shall not here discuss the detailed mechanisms of chemical bond formation, but we can simply look on the phenomenon as a balancing of forces. On the one hand there is a repulsion between the positively charged nuclei of both atoms, and between their negative electron 'clouds'; on the other there is an attraction between the nucleus of one atom and the electrons of the other, and vice versa. The two atoms settle at a mean internuclear distance such that these forces are just balanced and the total energy of the whole system is a minimum. Squeezing the atoms more closely together will cause the repulsive force to rise rapidly, while pulling them apart is resisted by the attractive force. Any attempt to distort the bond length requires an input of energy and we may plot energy against internuclear distance as in Fig. 3.1, where we have 'anchored' the chlorine atom on one axis and imagine pushing and pulling the hydrogen atom closer to or further from the chlorine—a bigger push or pull results in raising the energy more. At the energy minimum the internuclear distance is referred to as the equilibrium distance $r_{eq.}$, or more simply as the bond length.

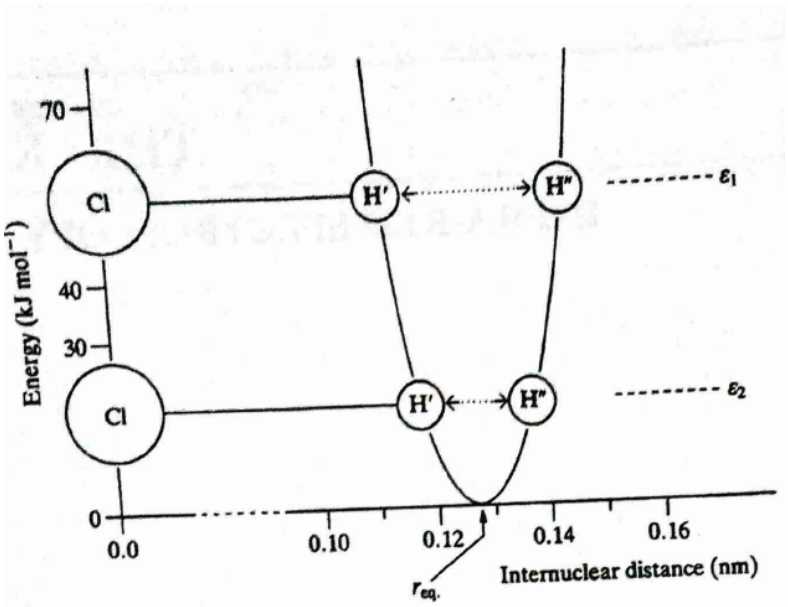


Figure 3.1 The energy of HCl as the bond is compressed or extended.

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The compression and extension of a bond may be likened to the behaviour of a spring, and we may extend the analogy by assuming that the bond, like a spring, obeys Hooke's law. We may write:

$$f = -k(r - r_{eq.}) \tag{3.1}$$

where f is the restoring force, k the force constant, and r the internuclear distance. In this case the energy curve is parabolic and has the form

$$E = \tfrac{1}{2}\, k(r - r_{eq.})^2 \tag{3.2}$$

This model of a vibrating diatomic molecule—the so-called simple harmonic oscillator model—while only an approximation, forms an excellent starting point for the discussion of vibrational spectra.

In Fig. 3.1 we have plotted the energy in the form of Eq. (3.2), i.e. as a parabola. The zero of curve and equation is at $r = r_{eq.}$, and any energy in excess of this, for example at ε_2, arises because of extension or compression of the bond. The figure shows that if one atom (Cl) is considered to be stationary on the $r = 0$ axis, the other (H) will oscillate between H′ and H″. In the case of HCl, it is a good approximation to say that, during vibrations, the heavy chlorine atom stays virtually still and it is the much lighter hydrogen which moves. However, only the *distance* between the two atoms is important and for any diatomic molecule we can always imagine ourselves to be sitting on one atom and watching the other move—from our point of view the atom we are on is stationary and can be assumed fixed on the $r = 0$ axis. Thus diagrams like Fig. 3.1 apply to *any* diatomic molecule.

If the energy of the HCl molecule of Fig. 3.1 is increased to ε_1 the oscillation will become more vigorous—that is to say the degree of compression and extension will be greater—but the vibrational frequency will not change. An elastic bond, like a spring, has a certain intrinsic vibrational frequency, dependent on the mass of the system and the force constant, but is independent of the amount of distortion. Classically it is simple to show that the oscillation frequency is:

$$\omega_{osc.} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}} \quad \text{Hz} \tag{3.3}$$

where μ is the reduced mass of the system (cf. Eq. (2.9)). To convert this frequency to wavenumbers, the unit most usually employed in vibrational spectroscopy, we must divide by the velocity of light, c, expressed in cm s^{-1} (cf. Sec. 1.1), obtaining:

$$\bar{\omega}_{osc.} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}} \quad \text{cm}^{-1} \tag{3.4}$$

Vibrational energies, like all other molecular energies, are quantized, and the allowed vibrational energies for any particular system may be calculated from the Schrödinger equation. For the simple harmonic oscillator these turn out to be:

$$E_v = (v + \tfrac{1}{2})h\omega_{osc.} \quad \text{joules} \quad (v = 0, 1, 2, \ldots) \tag{3.5}$$

where v is called the *vibrational quantum number*. Converting to the spectroscopic units, cm^{-1}, we have:

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$$\varepsilon_v = \frac{E_v}{hc} = (v + \tfrac{1}{2})\bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \tag{3.6}$$

as the only energies allowed to a simple harmonic vibrator. Some of these are shown in Fig. 3.2.

In particular, we should notice that the *lowest* vibrational energy, obtained by putting $v = 0$ in Eq. (3.5) or (3.6), is

$$E_0 = \tfrac{1}{2}h\omega_{\text{osc.}} \quad \text{joules} \quad (\omega_{\text{osc.}} \text{ in Hz})$$

or

$$\varepsilon_0 = \tfrac{1}{2}\bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \quad (\bar{\omega}_{\text{osc.}} \text{ in cm}^{-1}) \tag{3.7}$$

The implication is that the diatomic molecule (and, indeed, *any* molecule) can never have zero vibrational energy; the atoms can never be completely at rest relative to each other. The quantity $\frac{1}{2}h\omega_{\text{osc.}}$ joules or $\frac{1}{2}\bar{\omega}_{\text{osc.}}$ cm^{-1} is known as the zero-point energy; it depends only on the classical vibration frequency and hence (Eq. (3.3) or (3.4)) on the strength of the chemical bond and the atomic masses.

The prediction of zero-point energy is the basic difference between the wave mechanical and classical approaches to molecular vibrations. Classical mechanics could find no objection to a molecule possessing no vibrational energy but wave mechanics insists that it must always vibrate to some extent; the latter conclusion has been amply borne out by experiment.

Further use of the Schrödinger equation leads to the simple *selection rule* for the harmonic oscillator undergoing vibrational changes:

$$\Delta v = \pm 1 \tag{3.8}$$

To this we must, of course, add the condition that vibrational energy changes will only give rise to an observable spectrum if the vibration can interact with radiation i.e. (cf. Chapter 1) if the vibration involves a change in the dipole moment of the molecule. Thus vibrational spectra will be observable only in heteronuclear diatomic molecules since homonuclear molecules have no dipole moment.

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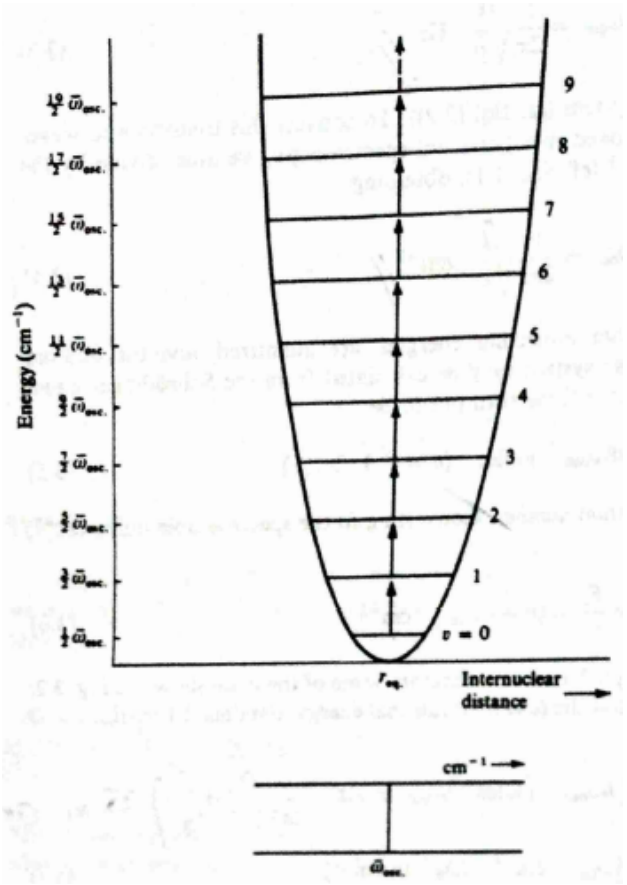


Figure 3.2 The vibrational energy levels and allowed transitions between them for a diatomic molecule undergoing simple harmonic motion.

Applying the selection rule we have immediately:

$$\varepsilon_{v+1\to v} = (v + 1 + \tfrac{1}{2})\bar{\omega}_{\text{osc.}} - (v + \tfrac{1}{2})\bar{\omega}_{\text{osc.}} = \bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \tag{3.9a}$$

for emission and

$$\varepsilon_{v\to v+1} = \bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \tag{3.9b}$$

for absorption, whatever the initial value of v.

Such a simple result is also obvious from Fig. 3.2—since the vibrational levels are equally spaced, transitions between any two neighbouring states will give rise to the same energy change. Further, since the difference between energy levels expressed in cm^{-1} gives directly the wavenumber of the spectral line absorbed or emitted

$$\bar{\nu}_{\text{spectroscopic}} = \varepsilon = \bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \tag{3.10}$$

This, again, is obvious if one considers the mechanism of absorption or emission in classical terms. In absorption, for instance, the vibrating molecule will absorb energy only from radiation with which it can coherently interact (cf. Fig. 1.8) and this must be radiation of its own oscillation frequency.

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21. **Explain in Vibrational-rotational spectra for heteronuclear diatomic molecule.**

25.4 VIBRATION-ROTATION SPECTRA

Molecules do not have a pure vibration spectrum due to the accompanying rotational transitions. This is shown by the fact that the selection rule ($\Delta v = \pm 1$) governing vibrational transitions is not obeyed at higher energy levels. As a result of this, the infrared region of spectrum is the vibration-rotation spectrum. The appearance of bands in the infrared region can thus be interpreted by supposing that the vibrational and rotational energies are additive and the energy of a linear molecule can be expressed as

$$E_{vib-rot} = E_{vib} + E_{rot}$$

$$= \left(v + \frac{1}{2}\right)h\nu_0 + BchJ(J+1)$$

For anharmonic oscillator and non-rigid rotator

$$E_{vib-rot} = \left(v + \frac{1}{2}\right)h\nu_0 \mp h\nu_0 x_e \left(v + \frac{1}{2}\right)^2 + BchJ(J+1) - DchJ^2[J+1]^2 + \ldots$$

where D is the centrifugal distortion constant and is given $D = \frac{4B^2}{\omega^2}$.

In the transition from energy level E to energy level E',

$$(\Delta E)_{v-j} = (E' - E)_{\text{vib-rot}}$$

$$= \left[\left(v' + \frac{1}{2}\right)h\nu_0 + BchJ'(J'+1)\right] - \left[\left(v + \frac{1}{2}\right)h\nu_0 + BchJ(J+1)\right]$$

$$= \left[(v' - v)h\nu_0\right] + \left[J'(J'+1) - J(J+1)\right]Bch$$

The selection rule for vibrational transition is $\Delta v = +1$, hence we have

$$\Delta E_{v-j} = h\nu_0 + \left[J'(J'+1) - J(J+1)\right]Bch$$

or
$$\nu = \frac{\Delta E_{v-j}}{h} = \nu_0 + \left[J'(J'+1) - J(J+1)\right]Bc$$

and the wave number is given as

$$\bar{\nu} = \frac{\nu}{c} = \frac{\Delta E_{v-j}}{hc} = \bar{\nu}_0 + \left[J'(J'+1) - J(J+1)\right]B \quad \ldots(25.19)$$

The selection rule for rotational transition is

$$\Delta J = \pm 1 \ (J' = J+1 \text{ or } J' = J-1)$$

The transitions for which $\Delta J = -1$ ($J' = J - 1$) produce the lines denoted as P branch of the spectrum (Fig. 25.7). The wave number of these lines is given by

$$\bar{\nu}_P = \bar{\nu}_0 - 2BJ \quad \ldots(25.20)$$

where J is the rotational quantum number of the initial state and have values of 1, 2, 3, ... etc.

The transitions which produce a set of lines arising from $\Delta J = 1$ ($J' = J + 1$) constitute the R branch of the spectrum as shown in Fig. 25.7. The wave number of the lines in R branch is given by

$$\bar{\nu}_R = \bar{\nu}_0 + 2B + 2BJ \quad \ldots(25.21)$$

where
$$J = 0, 1, 2, \ldots$$

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Equations (25.20) and (25.21) clearly show that the vibration-rotation bands are made up of two sets of lines P and R ($J \neq 0$) and there will not be a line with the fundamental vibration frequency. Therefore the actual origin of the band is marked by a gap. The lines so obtained would be equally spaced and the adjacent lines on either side would be separated by a distance $2B$.

Since $\Delta J = 0$ is not allowed, the transitions with $\Delta E_{v, J} = h\nu_0$ is not observed under these conditions. This is referred to as Q branch (Fig. 25.7). This appears as the origin of the band. The infra red spectrum of a diatomic molecule with this model will have two rotational vibrational band.

Equations (25.20) and (25.21) indicate that the spacing of lines in the P and R branches are equal to $2B$, *i.e.*, same as in pure rotational spectra. Hence from vibration-rotation spectra, the moment of inertia and from it the internuclear bond distance can be calculated. The fundamental vibration frequency ν_0 is obtained from the centre of the fundamental absorption band, *i.e.*, $\nu = 0$ to $\nu' = 1$. The fundamental vibration frequency can then be used to calculate the force constant (k) which is a measure of the stiffness of the bond.

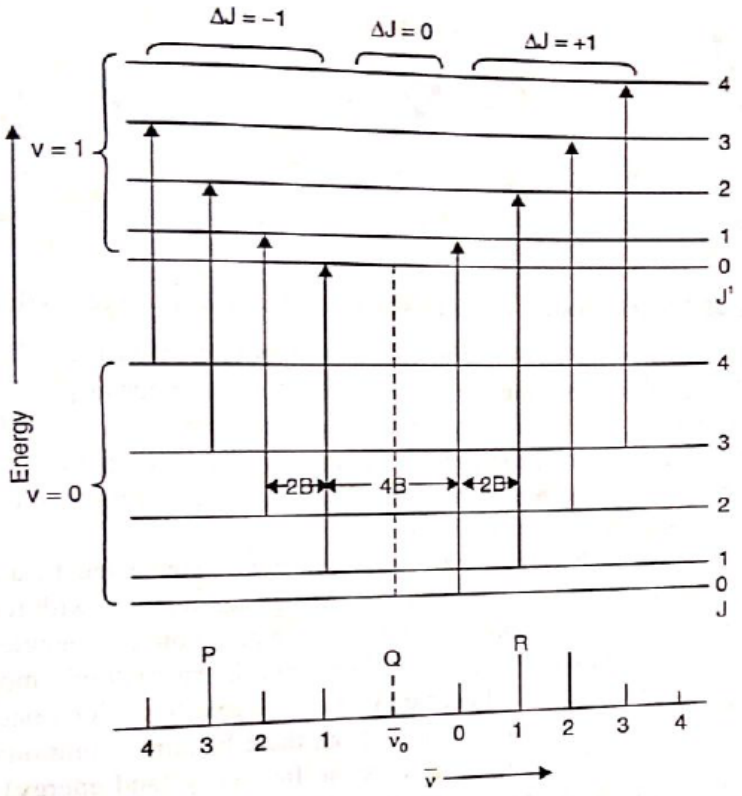


Fig. 25.7 Schematic energy levels showing structure of P, Q and R branches

22. What is the selection rule for IR spectroscopy?

The selection rule for a harmonic vibrational transition, according to the solution of Schrödinger equation is $\Delta v = \pm 1$. This means that the decrease or increase in the vibrational quantum number is by one unit only.

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vibrational levels involved in a transition are equally spaced as shown in Fig. 25.6. energy difference (ΔE) between two

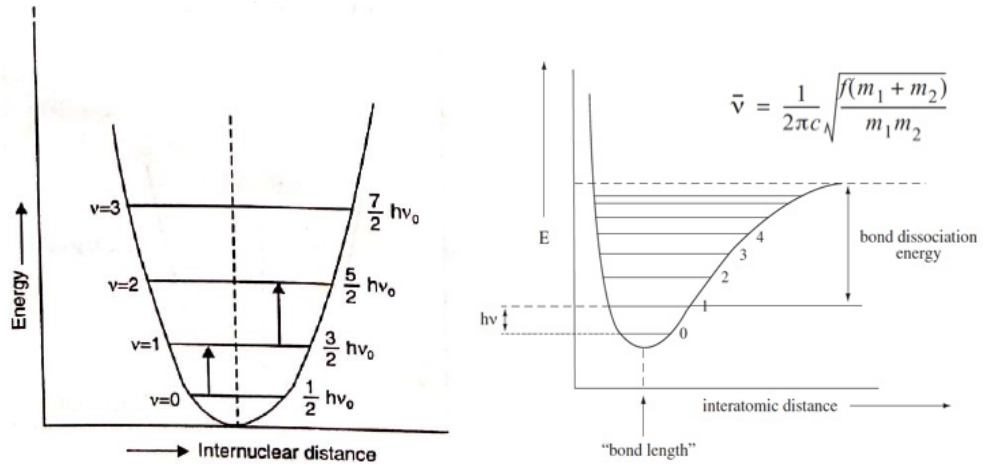
From equation (25.18) we have

$$\Delta E = h\nu_0$$

or $$h\nu = h\nu_0$$

or $$\nu = \nu_0$$

This shows that the frequency of absorbed radiation is the same as the fundamental frequency of vibration of the molecule. If follows that the vibrational spectrum should consist of a single line provided that there are no rotational energy changes. However, the vibrational spectra consists of a number of bands each with a fine structure of closely spaced lines. The presence of a number of bands is due to the fact that in the lowest state of vibration the diatomic molecule behaves as a simple harmonic oscillator, but at higher quantum levels there is a deviation from simple harmonic oscillator and the vibration becomes anharmonic. Because of this anharmonicity, the vibrational transitions do not confine to adjacent levels but also occur at different levels.



(a) (b)

Vibrational energy levels of (a) Harmonic oscillator (b) anharmonic oscillator

In anharmonic oscillator, the vibrational energy levels are more closely arranged at higher vibrational numbers. Therefore the selection rule for anharmonic oscillator is $\Delta v = \pm 1, \pm 2$ and ± 3.

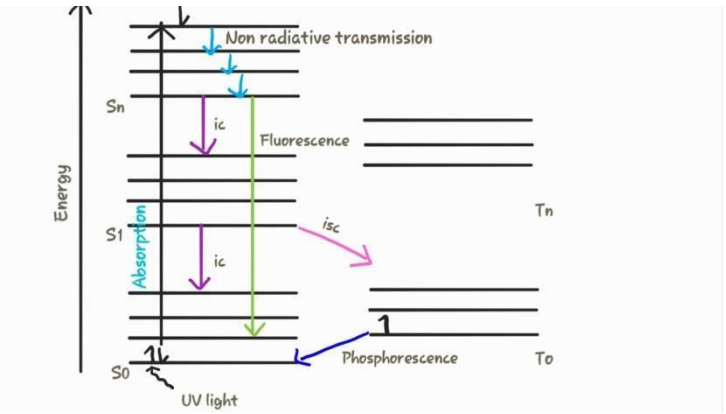
When $\Delta v = \pm 1$, it is called fundamental vibration. When $\Delta v = \pm 2$ or ± 3, is called overtones (first overtone or second overtone respectively).

23. **Explain the theory of Electronic spectroscopy**.

In electronic spectroscopy, we will be concerned with the absorption of light to bring the molecule to an excited electronic state.

Jablonski diagram explains the electronic transitions.

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Absorbance of a substance is explained by Beer-Lambert law. It states that Absorbance is directly proportional to the path length, *b*, and the concentration, *c*, of the absorbing species.

$A = \varepsilon bc$, where ε is a constant of proportionality, called the *absorbtivity*.

Electronic transitions are always accompanied by vibrational transitions.

A very useful guiding principle for investigating the vibrational structure of electron spectra is provided by the well known **Franck–Condon principle** which states that *electronic transition takes place so rapidly that a vibrating molecule does not change internuclear distance appreciably during the transition.* This principle is, to a first approximation true since the electrons move so much faster than the nuclei that during the electron transition the nuclei do not change their position. Hence, *an electronic transition may be represented by a vertical line on a plot of potential energy versus the internuclear distance.*

Let us demonstrate the Franck–Condon principle for the electronic transition of a diatomic molecule. Consider Fig. 23. where we have shown two potential energy curves for the molecule in the ground electronic state (E_0) and in the first excited electronic state (E_1). Since the bonding in the excited state is weaker than in the ground state, the minimum in the potential energy curve for the excited state occurs at a slightly greater internuclear distance than the corresponding minimum in the ground electronic state. Also, quantum mechanically it is known that the molecule is in the *centre* of the ground vibrational level of the ground electronic state. Thus, when a photon falls on the molecule, the most probable electronic transition, according to the Franck–Condon principle, takes place from $v''=0$ to $v'=2$ (written schematically as $0 \rightarrow 2$). Transitions to other vibrational levels of the excited electronic state occur with lower probabilities so that their relative intensities are smaller than the intensity of the $0 \rightarrow 2$ transition, as shown in Fig. 24.

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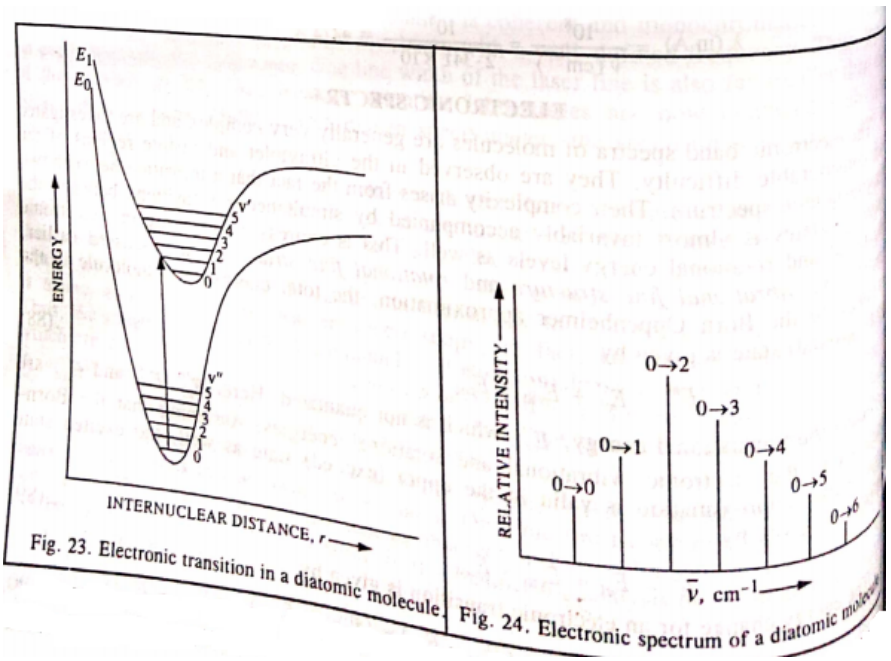


Fig. 23. Electronic transition in a diatomic molecule. Fig. 24. Electronic spectrum of a diatomic molecule

24. Identify from the following compounds, the type of electronic transitions it undergoes:

a. CH_2= CH_2 b. Benzene c. CH_3Cl d. CH_3 - C ≡ CH e. Cyclobutane

f. CH_2=CCl-CH=CH_2 g. Phenol h. Aniline

The possible electronic transitions are, **σ → σ*, *n* → σ*, *n* → π* and π → π* Transitions**

σ → σ* Transitions

An electron in a bonding σ orbital is excited to the corresponding antibonding orbital. The energy required is large. For example, methane (or all saturated molecules, which has only C-H bonds, and can only undergo σ → σ*transitions) shows an absorbance maximum at 125 nm. Absorption maxima due to σ → σ* transitions are not seen in typical UV-Vis. spectra (200 - 700 nm)

***n* → σ* Transitions**

Saturated compounds containing atoms with lone pairs (non-bonding electrons) are capable of *n* → σ* transitions. These transitions usually need less energy than σ → σ* transitions.

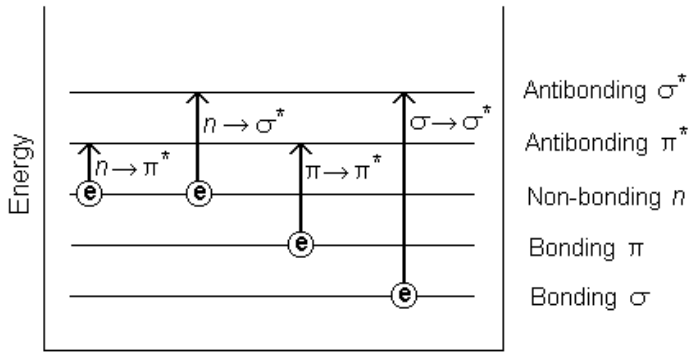
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They can be initiated by light whose wavelength is in the range 150 - 250 nm. The number of organic functional groups with $n \rightarrow \sigma^*$ peaks in the UV region is small.

$n \rightarrow \pi^*$ and $\pi \rightarrow \pi^*$ Transitions

Most absorption spectroscopy of organic compounds is based on transitions of n or π electrons to the π^* excited state. This is because the absorption peaks for these transitions fall in an experimentally convenient region of the spectrum (200 - 700 nm). These transitions need an unsaturated group in the molecule to provide the π electrons.

Molar absorbtivities from $n \rightarrow \pi^*$ transitions are relatively low, and range from 10 to100 L mol^{-1} $cm^{-1,}$ are less intensity of absorbance. $\pi \rightarrow \pi^*$ transitions normally give molar absorbtivities between 1000 and 10,000 L mol^{-1} cm^{-1}, therefore are more intensity of absorbance.



CH_2=CH_2 : Contains both sigma and pi bond and no n electrons. Therefore it can undergo $\pi \rightarrow \pi^*$ and $\sigma \rightarrow \sigma^*$ transitions. As $\sigma \rightarrow \sigma^*$ transitions cannot be studied, ethylene gives only $\pi \rightarrow \pi^*$ transitions.

Benzene: Contains both sigma and pi bond no n electrons. Therefore it can undergo $\pi \rightarrow \pi^*$ and $\sigma \rightarrow \sigma^*$ transitions. As $\sigma \rightarrow \sigma^*$ transitions cannot be studied, gives only $\pi \rightarrow \pi^*$ transitions.

CH_3Cl : Contains sigma and n electrons. Therefore it can undergo only $\sigma \rightarrow \sigma^*$ and $n \rightarrow \sigma^*$ transition. As $\sigma \rightarrow \sigma^*$ transitions cannot be studied, it gives only $n \rightarrow \sigma^*$ transition in UV-VIS spectra.

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CH_3 - C ≡ CH : Contains both sigma and pi bond and no n electrons. Therefore it can undergo $\pi \rightarrow \pi^*$ and $\sigma \rightarrow \sigma^*$ transitions. As $\sigma \rightarrow \sigma^*$ transitions cannot be studied, gives only $\pi \rightarrow \pi^*$ transitions.

Cyclobutane : Contains only sigma electrons. Therefore it can undergo only $\sigma \rightarrow \sigma^*$ transition. As $\sigma \rightarrow \sigma^*$ transitions cannot be studied, it does not give UV-VIS spectra.

CH_2=CCl-CH=CH_2: Contains both sigma, pi and n electrons. Therefore it can undergo all $n \rightarrow \pi^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \sigma^*$ and $\sigma \rightarrow \sigma^*$ transitions. As $\sigma \rightarrow \sigma^*$ transitions cannot be studied, this gives $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ and $n \rightarrow \sigma^*$ transitions.

Phenol : Contains both sigma, pi and n electrons. Therefore it can undergo all $n \rightarrow \pi^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \sigma^*$ and $\sigma \rightarrow \sigma^*$ transitions. As $\sigma \rightarrow \sigma^*$ transitions cannot be studied, this gives $n \rightarrow \pi^*$, $\pi \rightarrow \pi^*$ and $n \rightarrow \sigma^*$ transitions.

Aniline : Contains both sigma and n electrons. Therefore it can undergo $n \rightarrow \sigma^*$ and $\sigma \rightarrow \sigma^*$ transitions. As $\sigma \rightarrow \sigma^*$ transitions cannot be studied, this gives $n \rightarrow \sigma^*$ transitions.

25. What is spin selection rule in electronic spectra? Explain.

26. **Explain Laporte or orbital selection rule**.

27. **Discuss the selection rule for H atom in electronic spectroscopy**.

28. Explain the principle behind the absorption of radio frequency waves. [Or] What are the criteria for a molecule to absorb NMR radiation?

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Atomic nuclei possessing nuclear spin (I > 0) are NMR active. eg., ^{1}H, ^{19}F, etc.

Atomic nuclei does not possessing nuclear spin (I = 0) are NMR in-active. eg., $^{4}He_{2}$, $^{12}C_{6}$, $^{16}O_{8}$.

All atomic nuclei possess nuclear spin, I, which may be integral (*i.e.*, 1, 2, 3, etc.) or half-integral (*i.e.*, 1/2, 3/2, 5/2, 7/2 etc.). The proton and the neutron each has spin 1/2. Since a nucleus contains nucleons (collective term for protons and neutrons), it is evident that the spin of the nucleus may be considered as the resultant of the spins of protons and neutrons comprising it.

Since a nucleus possesses an electric charge, *the spinning nucleus gives rise to a magnetic field whose axis coincides with the axis of spin*. Thus, each nucleus can be thought of being equivalent to a minute magnet having a magnetic moment μ. Each nucleus with $I > 0$ has magnetic moment. However, $^{12}_{6}C$ and $^{16}_{8}O$ having $I=0$ do not possess a magnetic moment, i.e., they are non-magnetic.

When a magnetic nucleus is placed in a uniform magnetic field, the magnetic ... associated with the magnetic moment assumes only *a discrete set of orientations*. ... expressed by saying that the nuclear spin (and hence the magnetic moment associated ... is *quantized* with respect to the direction of the applied magnetic field. The magnetic ... assumes any one of the $2I+1$ orientations with respect to the direction of the applied ... field. Suppose that a nucleus with spin I is placed in an external magnetic field B_z ... along the z-direction. Then, as a result of quantization of spin, the observable compone... of the spin I along the direction of the field is given by

(i) $m_I = I, I-1, I-2, \ldots, 0, \ldots, -(I-2), -(I-1), -I$ (*for integral spins*)

(ii) $m_I = I, I-1, I-2, \ldots, 1/2, -1/2, -3/2, \ldots, -I$ (*for half-integral spins*)

In each case, there are $(2I+1)$ values of m_I.

As mentioned earlier, a nucleus with spin has associated with it a magnetic ... moment (or simply magnetic moment) μ that is proportional to the angular momenta... whose *magnitude* is given by $(I(I+1)^{1/2}\ (h/2\pi)$. Thus,

$$\vec{\mu} = g_N \mu_N \vec{I}$$

where g_N is called the nuclear g-factor and μ_N is the nuclear megneton defin... $\mu_N = eh/4\pi m_p$, where e is the electronic charge, m_p is the mass of the proton. Substitutin... various values,

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$$E = -\vec{\mu}\,\vec{B_z} \qquad \ldots(102)$$

Substituting the value of $\vec{\mu}$ from Eq. 100, we find that

$$E = -g_N \mu_N \vec{B_z}\,\vec{I} \qquad \ldots(103)$$

As already mentioned, the nuclear spin I is quantized in the presence of the magnetic field and the energy of nuclear spin is defined only by its component m_I. We may thus write :

$$E_{m_I} = -g_N \mu_N B_z m_I \qquad \ldots(104)$$

Consider a bare proton ^{1}H with $I=1/2$ so that $m_I = \pm 1/2$. Thus, for $m_I = 1/2$, the energy

$$E_{1/2} = -\tfrac{1}{2} g_N \mu_N B_z \qquad \ldots(105)$$

and for $m_I = -\frac{1}{2}$, the energy

$$E_{-1/2} = \tfrac{1}{2} g_N \mu_N B_z \qquad \ldots(106)$$

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It should be noted that in the absence of the external magnetic field, the two components of the proton spin have the same energy (*i.e.*, they are *degenerate*). However, in the presence of the magnetic field, the components have different energies given by Eqs. 105 and 106. This is expressed by saying that *the magnetic field lifts* or *removes the degeneracy of the spin components*. This is called *Zeeman splitting of energy levels*. The energy difference is given by

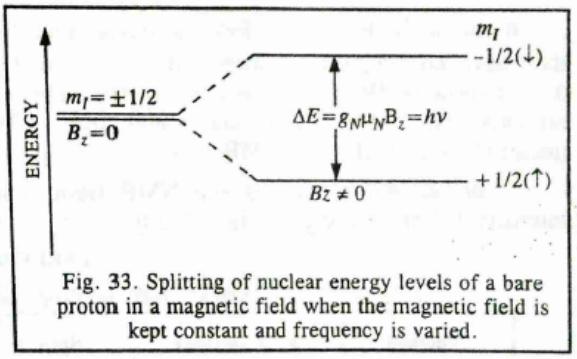


Fig. 33. Splitting of nuclear energy levels of a bare proton in a magnetic field when the magnetic field is kept constant and frequency is varied.

$$\Delta E = E_{-1/2} - E_{1/2} = g_N \mu_N B_z \qquad \ldots(107)$$

This is illustrated in Fig. 33.

In order to induce transitions between the two energy levels $E_{-1/2}$ and $E_{1/2}$, *i.e.*, in order to let the nuclear spin 'flip' from the upward direction (↑) to the downward direction (↓), we must apply an oscillating radiofrequency field perpendicular to the direction of B_z. *The nuclear spin flips, provided the energy* $h\nu$ of the radiofrequency photon of frequency ν is exactly equal to ΔE, *i.e.*,

$$\Delta E = g_N \mu_N B_z = h\nu \qquad \ldots(108)$$

This is called the **Bohr frequency condition** Thus, the NMR frequency of a **bare proton** (or a bare nucleus with $I = 1/2$) is given by

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It may be noted that g_N can be positive or negative but ν must be positive. Hence, only the absolute value of g_N should be used in calculating ν.

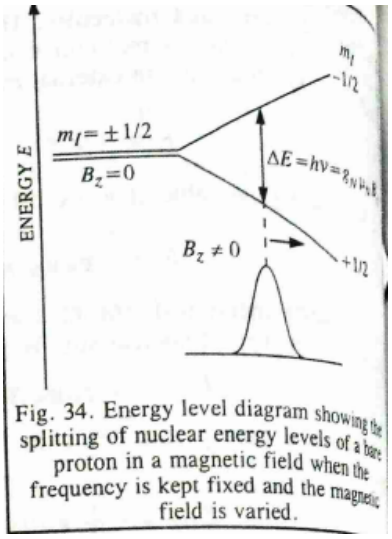


Fig. 34. Energy level diagram showing the splitting of nuclear energy levels of a bare proton in a magnetic field when the frequency is kept fixed and the magnetic field is varied.

29. **Define the term chemical shift and explain it taking ethanol as an example**.

(OR)

Explain how position of signals in NMR spectra is interpreted, taking an example.

(OR)

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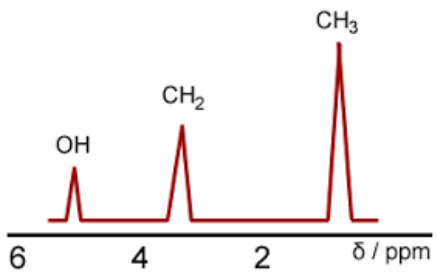
Explain in detail the influence of electronic environment on the position of signals taking an example in NMR spectroscopy

The Chemical Shift. In practice, one does not investigate the NMR spectra of 'bare' nuclei. Nuclei are essential parts of molecules held together by the electrons of the chemical bonds. The magnetic field at which a free or 'bare' nucleus resonates is quite different from the field at which the same nucleus in a molecule resonates. It is thus clear that the electrons in a molecule affect the NMR frequency of a given nucleus in the molecule. The electrons are said to *shield* or *deshield* a nucleus from the applied magnetic field, depending upon whether the magnetic field generated by their motion opposes or reinforces the field. We shall discuss this aspect a little later in this chapter. Suffice it to mention here that the applied external magnetic field affects the motion of the electrons surrounding the nuclei thereby inducing local magnetic fields. The magnitude of the local magnetic field is determined by the electron distribution around a nucleus. The difference between the magnitudes of the magnetic field at which 'free' nuclei and molecular nuclei resonate is called **chemical shift.**

Chemical shift δ is usually expressed in parts per million (ppm) by frequency, because it is calculated from:

$$\delta = \frac{\nu_{\text{sample}} - \nu_{\text{TMS}}}{\nu_0} \times 10^6 \text{ (ppm)}$$

where ν_{sample} is the absolute resonance frequency of the sample and ν_{TMS} is the absolute resonance frequency of a standard reference compound TMS, measured in the same applied magnetic field B_0. Since the numerator is usually expressed in hertz, and the denominator in megahertz, δ is expressed in ppm.



Consider the proton-NMR spectrum of ethanol (Fig. 37). Each proton in the pure compound has its own electronic environment and hence a characteristic chemical shift. The liquid ethanol shows in its NMR spectrum three peaks at three values of the applied magnetic field corresponding to the -OH proton, the two equivalent $>CH_2$ protons and the three equivalent $-CH_3$ protons. The areas under the three absorption peaks are in the ratio of 1 : 2 : 3, respectively

30. **Write notes on the theory behind NMR spectra**.

All atomic nuclei possess nuclear spin, I, which may be integral (*i.e.*, 1, 2, 3, etc.) or half-integral (*i.e.*, 1/2, 3/2, 5/2, 7/2 etc.). The proton and the neutron each has spin 1/2. Since a nucleus contains nucleons (collective term for protons and neutrons), it is evident that the spin of the nucleus may be considered as the resultant of the spins of protons and neutrons comprising it.

Since a nucleus possesses an electric charge, *the spinning nucleus gives rise to a magnetic field whose axis coincides with the axis of spin*. Thus, each nucleus can be thought of being equivalent to a minute magnet having a magnetic moment μ. Each nucleus with $I > 0$ has magnetic moment. However, $^{12}_{6}C$ and $^{16}_{8}O$ having $I=0$ do not possess a magnetic moment, i.e., they are non-magnetic.

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When a magnetic nucleus is placed in a uniform magnetic field, the magnetic
associated with the magnetic moment assumes only *a discrete set of orientations*.
expressed by saying that the nuclear spin (and hence the magnetic moment associated
is *quantized* with respect to the direction of the applied magnetic field. The magnetic
assumes any one of the $2I+1$ orientations with respect to the direction of the applied
field. Suppose that a nucleus with spin I is placed in an external magnetic field B_z
along the z-direction. Then, as a result of quantization of spin, the observable
of the spin I along the direction of the field is given by

(i) $m_I = I, I-1, I-2, \ldots, 0, \ldots, -(I-2), -(I-1), -I$ (*for integral spins*)

(ii) $m_I = I, I-1, I-2, \ldots, 1/2, -1/2, -3/2, \ldots, -I$ (*for half-integral spins*)

In each case, there are $(2I+1)$ values of m_I.

As mentioned earlier, a nucleus with spin has associated with it a magnetic
moment (or simply magnetic moment) μ that is proportional to the angular
whose *magnitude* is given by $(I(I+1)^{1/2}\ (h/2\pi)$. Thus,

$$\vec{\mu} = g_N \mu_N \vec{I}$$

where g_N is called the nuclear g-factor and μ_N is the nuclear megneton
$\mu_N = eh/4\pi m_p$, where e is the electronic charge, m_p is the mass of the proton. Substitu
various values,

NMR of a 'bare' nucleus. A 'bare' nucleus is, of course, an ideality. Actually, nuclei are parts of atoms and molecules. However, it is useful to consider the NMR of a bare nucleus. From quantum mechanics it is known that the energy of interaction of nuclear magnetic moment μ with an external magnetic field B_z, applied in the z-direction, is given by

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$$E_{1/2} = -\tfrac{1}{2} g_N \mu_N B_z \qquad \ldots(105)$$

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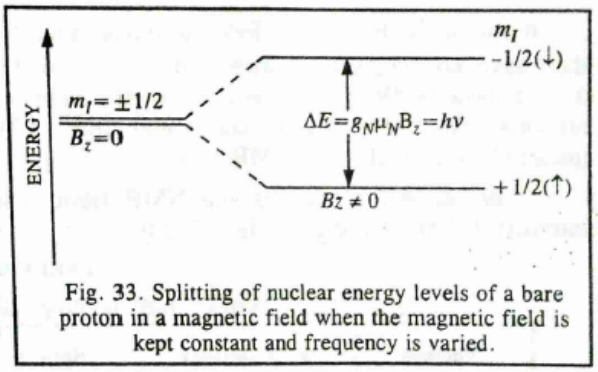


Fig. 33. Splitting of nuclear energy levels of a bare proton in a magnetic field when the magnetic field is kept constant and frequency is varied.

$$\Delta E = E_{-1/2} - E_{1/2} = g_N \mu_N B_z \qquad \ldots(107)$$

This is illustrated in Fig. 33.

In order to induce transitions between the two energy levels $E_{-1/2}$ and $E_{1/2}$, *i.e.*, in order to let the nuclear spin 'flip' from the upward direction ($\uparrow$) to the downward direction ($\downarrow$), we must apply an oscillating radiofrequency field perpendicular to the direction of B_z. *The nuclear spin flips, provided the energy* $h\nu$ of the radiofrequency photon of frequency ν is exactly equal to ΔE, *i.e.*,

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$$\nu = \frac{\Delta E}{h} = \frac{g_N \mu_N B_z}{h} \qquad \ldots(109)$$

It may be noted that g_N can be positive or negative but ν must be positive. Hence, only the absolute value of g_N should be used in calculating ν.

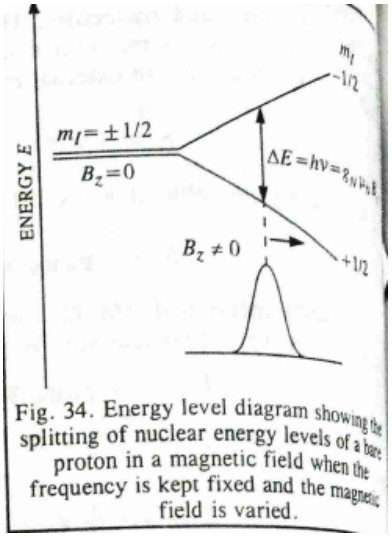


Fig. 34. Energy level diagram showing the splitting of nuclear energy levels of a bare proton in a magnetic field when the frequency is kept fixed and the magnetic field is varied.

Eq. 109 shows that ν is directly proportional to B_z. We thus see that the NMR spectra can be recorded in two ways, *viz.*, *(i)* keeping the magnetic field B_z fixed and varying the frequency ν (Fig. 33) or *(ii)* keeping ν fixed and varying B_z (Fig. 34). The latter mode is preferred in modern NMR spectrometers.

31. **What are NMR active and inactive nuclei? Give examples**.

Atomic nuclei possessing nuclear spin (I > 0) are NMR active. eg., ^{1}H, ^{19}F, etc.

Atomic nuclei does not possessing nuclear spin (I = 0) are NMR in-active. eg., $^{4}He_2$, $^{12}C_6$, $^{16}O_8$.

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All atomic nuclei possess nuclear spin, I, which may be integral (*i.e.*, 1, 2, 3, etc.) or half-integral (*i.e.*, 1/2, 3/2, 5/2, 7/2 etc.). The proton and the neutron each has spin 1/2. Since a nucleus contains nucleons (collective term for protons and neutrons), it is evident that the spin of the nucleus may be considered as the resultant of the spins of protons and neutrons comprising it.

It is difficult to predict the spin of a nucleus. The following rules are, however, helpful in predicting the nuclear spin :

(i) If the mass number A is odd, nuclear spin I is *half integral*. Thus, in the case of ^{1}H, ^{15}N, ^{19}F, ^{31}P, $I = 1/2$ and in the case of ^{11}B, $I = 3/2$.

(ii) If the mass number A and the atomic number Z are both even, the spin is *zero*. Thus, in the case of $^{4}_{2}He$, $^{12}_{6}C$, $^{16}_{8}O$, $I = 0$.

(iii) If the mass number A is even but the atomic number Z is odd, the spin is *integral*. Thus, the spin of $^{2}_{1}H$ and $^{14}_{7}N$ is 1 while that of $^{10}_{5}B$ is 3.

Since a nucleus possesses an electric charge, *the spinning nucleus gives rise to a magnetic field whose axis coincides with the axis of spin*. Thus, each nucleus can be thought of being equivalent to a minute magnet having a magnetic moment μ. Each nucleus with $I > 0$ has magnetic moment. However, $^{12}_{6}C$ and $^{16}_{8}O$ having $I=0$ do not possess a magnetic moment, i.e., they are non-magnetic.

32. **What are the two scales used to calculate chemical shift values? Give any two references (or) standards used in NMR spectroscopy**.

It is pertinent to mention here that two chemical-shift scales, *viz.*, the δ-scale and the τ-scale are used in recording NMR signals. They are related by the expression

$$\delta + \tau = 10$$

i.e.,
$$\tau = 10 - \delta \qquad \ldots(118)$$

On the δ-scale, δ is set equal to zero at the reference peak (or signal) of TMS and it *increases downfield*. This scale has the disadvantage that a large value of δ implies that a proton undergoes a resonance at low field. Hence, it implies a low-shielding of the nucleus by the external magnetic field.

In the τ-scale, τ is set equal to 10 at the reference peak of TMS and it *decreases downfield*. Thus, a low value of τ implies that a proton undergoes a resonance at low field. Hence, it implies a low-shielding of the nucleus by the external magnetic field. It follows that a higher value of τ means a more shielded nucleus and hence a high-field resonance.

In NMR spectroscopy. Tetramethylsilane is the accepted internal standard for calibrating chemical shift for ^{1}H, ^{13}C and ^{29}Si NMR spectroscopy in organic solvents (where TMS is soluble). In water, where it is not soluble, **sodium** salts of DSS, 2,2-dimethyl-2-silapentane-5-sulfonate, are used instead.

33. What is shielding and de-shielding effect in NMR spectra?

Shielding and Deshielding of Protons. The concept of chemical shift, introduced above, is closely linked with the shielding or deshielding of protons in a molecule. A *bare* proton, when exposed to a radiofrequency photon, 'flips' its spin from lower energy level to a higher energy level, *i.e.*, its spin changes its orientation from (↑) to (↓). However, all protons do not change or flip their spins at the same applied magnetic field because their energy depends upon the chemical environment of the protons. The magnetic field 'sensed' by a proton in a molecule is not the same as the applied magnetic field since the electrons in the bond close to the proton and the neighbouring electrons in other bonds *induce* their own magnetic field

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This induced field may oppose or reinforce the applied magnetic field. The field experienced by the proton is thus the resultant of the applied and induced fields, as shown in Fig. 38. When the two fields reinforce, as in Fig. 38*(b)*, a smaller field must be applied to flip the proton. Such a proton is said to be *deshielded* and absorbs more *down-field* (*i.e.*, at lower magnetic field strength). On the other hand, when the fields oppose, as in Fig. 38*(c)*, a stronger field must be applied. The proton is now *shielded* and absorbs more *upfield* (*i.e.*, at higher magnetic field strength).

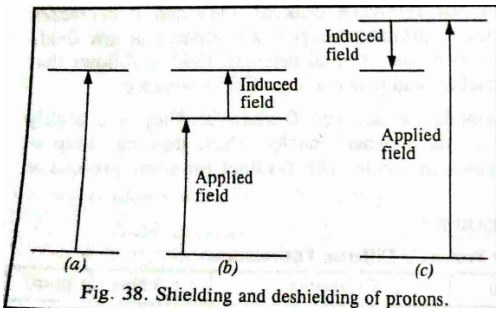


Fig. 38. Shielding and deshielding of protons.

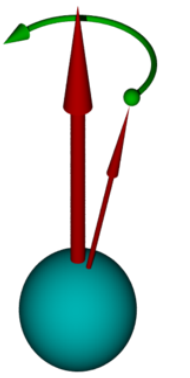
34. What is Larmor (or) precessional frequency?

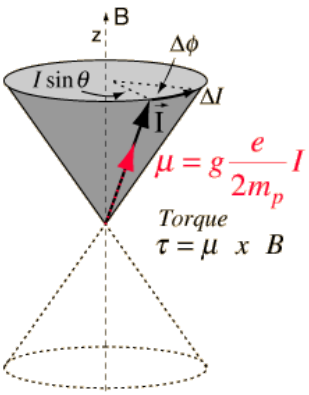
Larmor precession is the precession of the magnetic moment of an object about an external magnetic field.

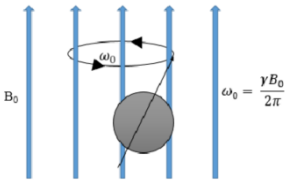
$$\omega = \gamma B_z \qquad \ldots(110)$$

where γ is the magnetogyric ratio (formerly called gyromagnetic ratio) of the nucleus. It is defined as the ratio of the nuclear magnetic moment μ and the nuclear spin angular momentum $I(h/2\pi)$. *i.e.*,

$$\gamma = \frac{\mu}{I(h/2\pi)}$$







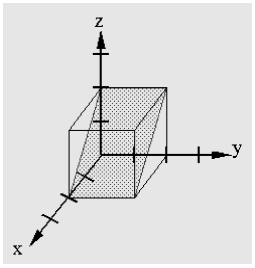
The large arrow indicates the external magnetic field, the small arrow the spin angular momentum of the particle.

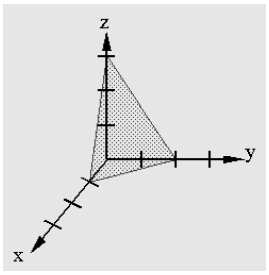
MODULE III

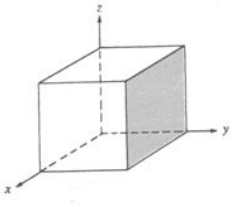
1. Explain the principle of XPS?

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2. Define Miller indices with examples.
3. For the intercepts x, y, and, z with values of 3, 1, and 2 respectively, find the Miller indices.
4. Compute the Miller Indices for a plane intersecting at x= ¼, y=1, and x=1/2.
5. Give the expression for Bragg's law and explain the terms involved in it.
6. What is inter-plane spacing's in lattices? Give the expression taking an example.
7. Determine the Miller indices (hkl) of the shaded planes below. Show your work on each step of Miller indices to determine the plane.



b.


c.


8. Discuss the principle, instrumentation and applications of XPS.
9. Explain Bragg's law with a neat diagram.
10. Define the terms i. Critical temperature ii. Critical volume iii. Critical pressure.
11. Write notes on dipole-dipole interactions.
12. What is modified form of Vander Waals equation?
13. Give the Clausius equation for real gases.

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